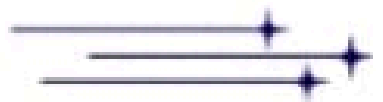
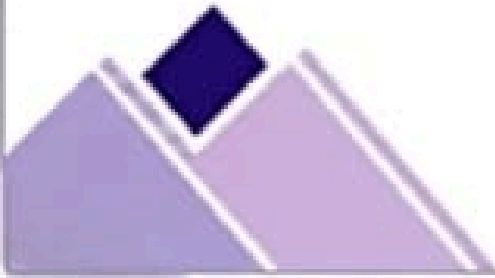


2.7

Polar and Euler Form of a Complex Number

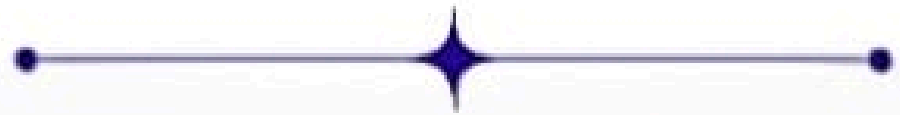
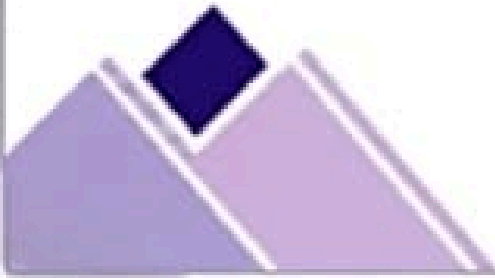
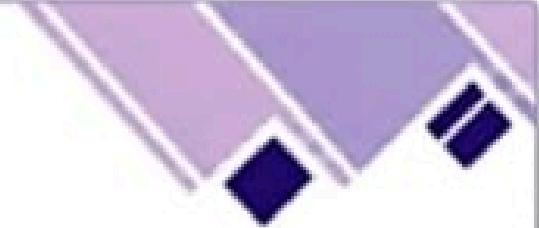
Step-by-step notes with all figures from Section 2.7 · XII Mathematics



Why do we need polar form?

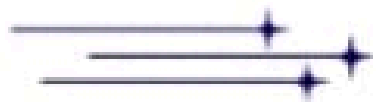
- ✦ Addition & subtraction of complex numbers → use rectangular form ($x + iy$). We simply add/subtract real and imaginary parts.
- ✦ Multiplication, powers, and roots of complex numbers → use polar form. Computations become much easier.
- ✦ Polar form describes a complex number using distance and direction from the origin, instead of x and y .

Rectangular form ($x + iy$) → best for $+$, $-$
Polar form (r, θ) → best for $\times$, powers, roots



2.7.1 Polar Form of a Complex Number

Setting up the polar coordinate system



Step 1 – The polar coordinate system

- ◆ O is a fixed point called the pole.
- ◆ The horizontal half-line from O is called the initial line (polar axis).
- ◆ r = distance from pole O to point P.
- ◆ θ = angle from the initial line to OP, measured counter-clockwise.
- ◆ (r, θ) together are called the polar coordinates of P.

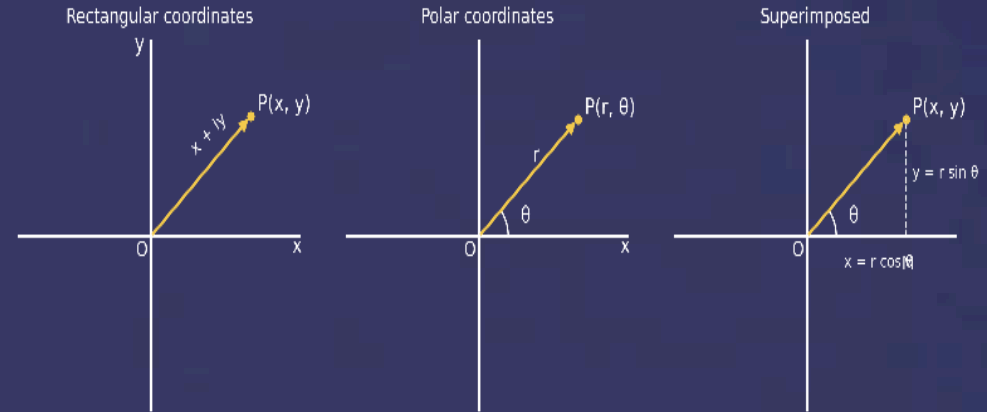


Fig 2.26 – 2.28 : Rectangular $\leftrightarrow$ Polar $\leftrightarrow$ Superimposed

Superimposing polar on rectangular coordinates gives:

$$x = r \cos \theta \quad \dots(1)$$

$$y = r \sin \theta \quad \dots(2)$$

So any non-zero $z = x + iy$ can be written as $z = r \cos \theta + i r \sin \theta$.



Step 2 — Definition 2.6: Polar form

$$z = r (\cos \theta + i \sin \theta)$$

the polar form (trigonometric form) of $z = x + iy$

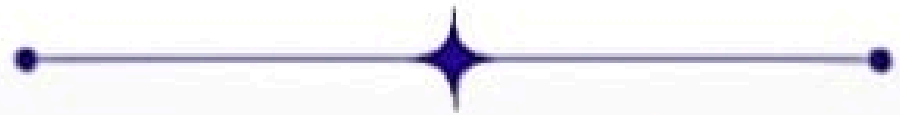
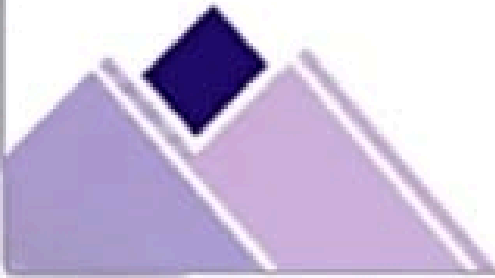
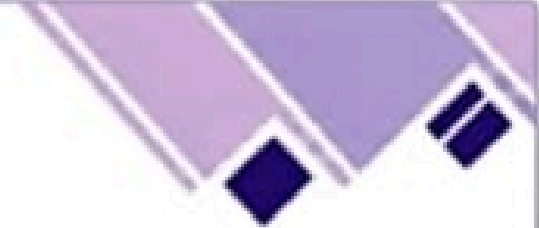
- ◆ Short notation: $z = r \operatorname{cis} \theta$ (short for $r \cos \theta + i r \sin \theta$).
- ◆ $r = |z|$ is called the absolute value or modulus of z .
- ◆ θ is called the argument (or amplitude) of z , written $\theta = \arg(z)$.

Step 3 — Notes on modulus and argument

- ◆ (i) If $z = 0$, the argument θ is undefined. So we always assume $z \neq 0$ when using polar coordinates.
- ◆ (ii) If $z = x + iy$ has polar coordinates (r, θ) , its conjugate $\bar{z} = x - iy$ has polar coordinates $(r, -\theta)$.

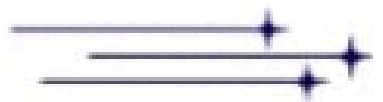
Squaring and adding (1) & (2): $r = |z| = \sqrt{x^2 + y^2}$

Dividing (2) by (1): $\tan \theta = y / x$



Two Kinds of Argument

General argument vs. Principal argument



Case (i): the general argument, $\arg z$

- ✦ θ is the angle (in radians) that z makes with the positive real axis, treating z as a radius vector.
- ✦ θ has infinitely many values — they differ from each other by integer multiples of 2π .
- ✦ All these values are found from $\tan \theta = y/x$, choosing the correct quadrant for the point z .
- ✦ The set of all such values is denoted $\arg z$.



Case (ii): the principal argument, $\text{Arg } z$

- ◆ There is exactly one value of θ satisfying $-\pi < \theta \leq \pi$.
- ◆ This unique value is called the principal value / principal argument of z , written $\text{Arg } z$ (capital A).
- ◆ The capital A distinguishes the single principal value from the general set $\arg z$.

$$-\pi < \text{Arg}(z) \leq \pi$$

General rule: $\arg z = \text{Arg } z + 2n\pi, n \in \mathbb{Z}$

Step 4 – Finding the principal angle by quadrant

- ◆ First compute the reference angle: $\alpha = \tan^{-1} |y/x|$.
- ◆ Then adjust α using π , depending on which quadrant z lies in:

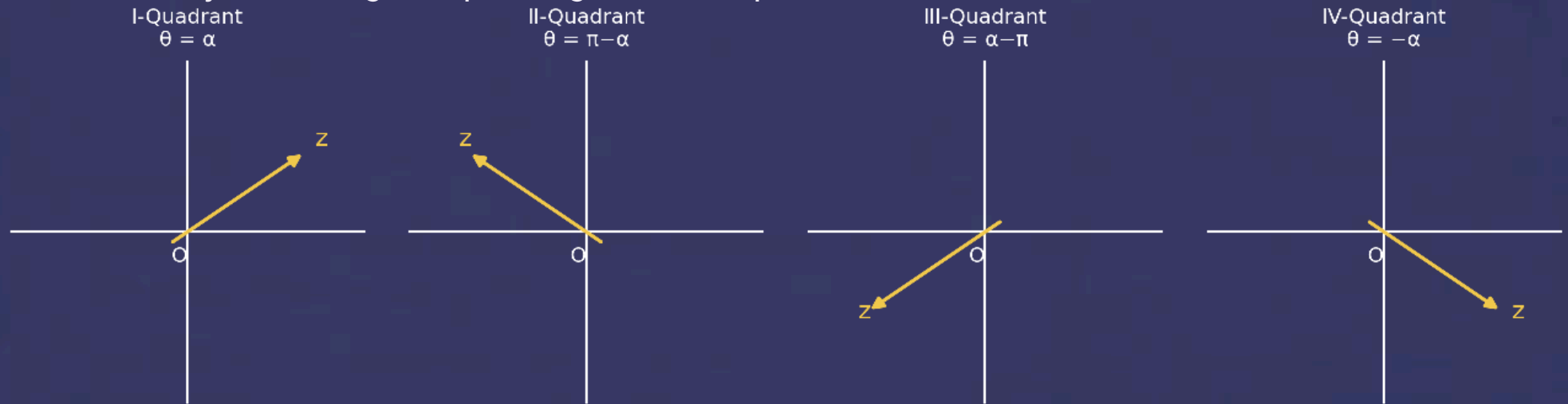


Fig 2.30 – 2.33 : Principal argument in each quadrant

$$\text{I: } \theta = \alpha \quad \text{II: } \theta = \pi - \alpha \quad \text{III: } \theta = \alpha - \pi \quad \text{IV: } \theta = -\alpha$$

Step 5 — Properties of arguments

- ◆ (1) $\arg(z_1 z_2) = \arg z_1 + \arg z_2$
- ◆ (2) $\arg(z_1 / z_2) = \arg z_1 - \arg z_2$
- ◆ (3) $\arg(z^n) = n \cdot \arg z$
- ◆ (4) $\cos \theta + i \sin \theta$ is the same as $\cos(2k\pi + \theta) + i \sin(2k\pi + \theta)$, $k \in \mathbb{Z}$



Example – Argument of 1, i, -1, -i

z	1	i	-1	$-i$
$\text{Arg}(z)$	0	$\pi/2$	π	$-\pi/2$
$\arg z$	$2n\pi$	$2n\pi + \pi/2$	$2n\pi + \pi$	$2n\pi - \pi/2$

These are the four points on the unit circle at 0, $\pi/2$, π , and $-\pi/2$ radians.

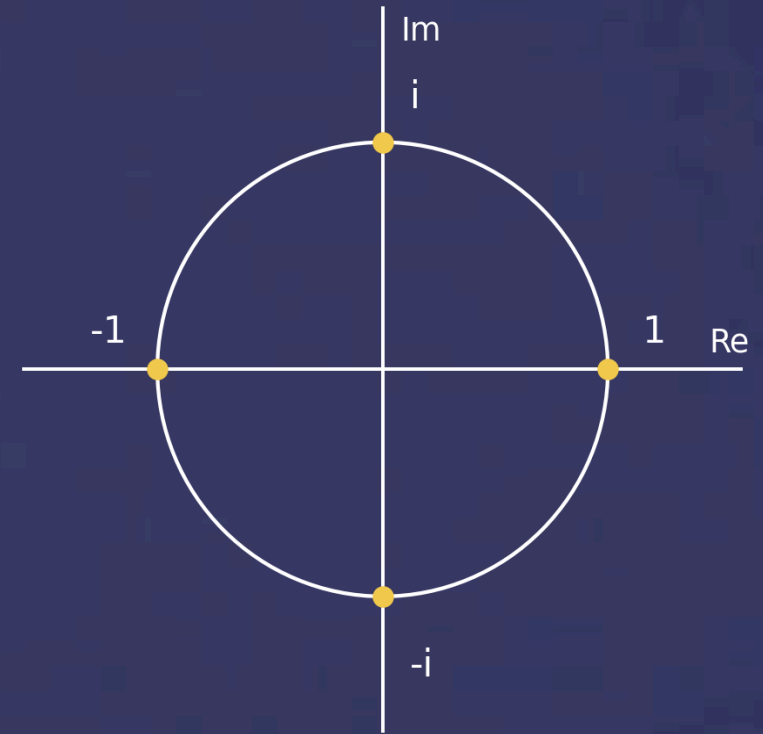
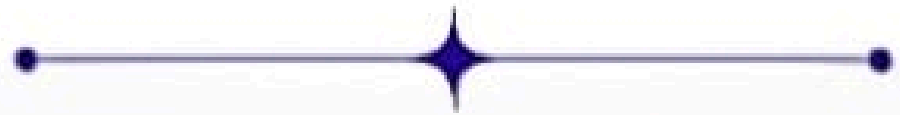
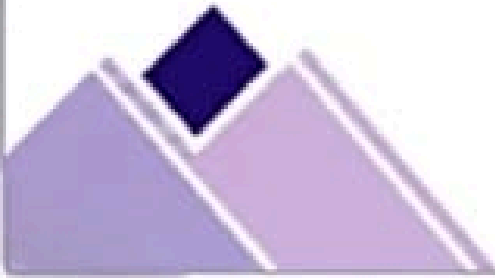
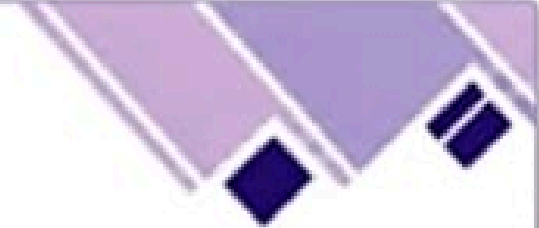
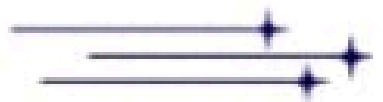


Fig 2.34 : Unit circle



Illustration

Plotting complex numbers given in polar form



Plot these complex numbers in the complex plane

- ◆ (i) $5 (\cos \pi/4 + i \sin \pi/4) = 5 \operatorname{cis}(\pi/4)$
- ◆ (ii) $4 (\cos 2\pi/3 + i \sin 2\pi/3) = 4 \operatorname{cis}(2\pi/3)$
- ◆ (iii) $3 (\cos(-5\pi/6) + i \sin(-5\pi/6)) = 3 \operatorname{cis}(-5\pi/6)$
- ◆ (iv) $2 (\cos \pi/6 - i \sin \pi/6) = 2 \operatorname{cis}(-\pi/6)$

Method: for each, r is the distance from O , and θ (from the $\cos/\sin$ argument) is the angle from the positive real axis — negative θ goes clockwise.

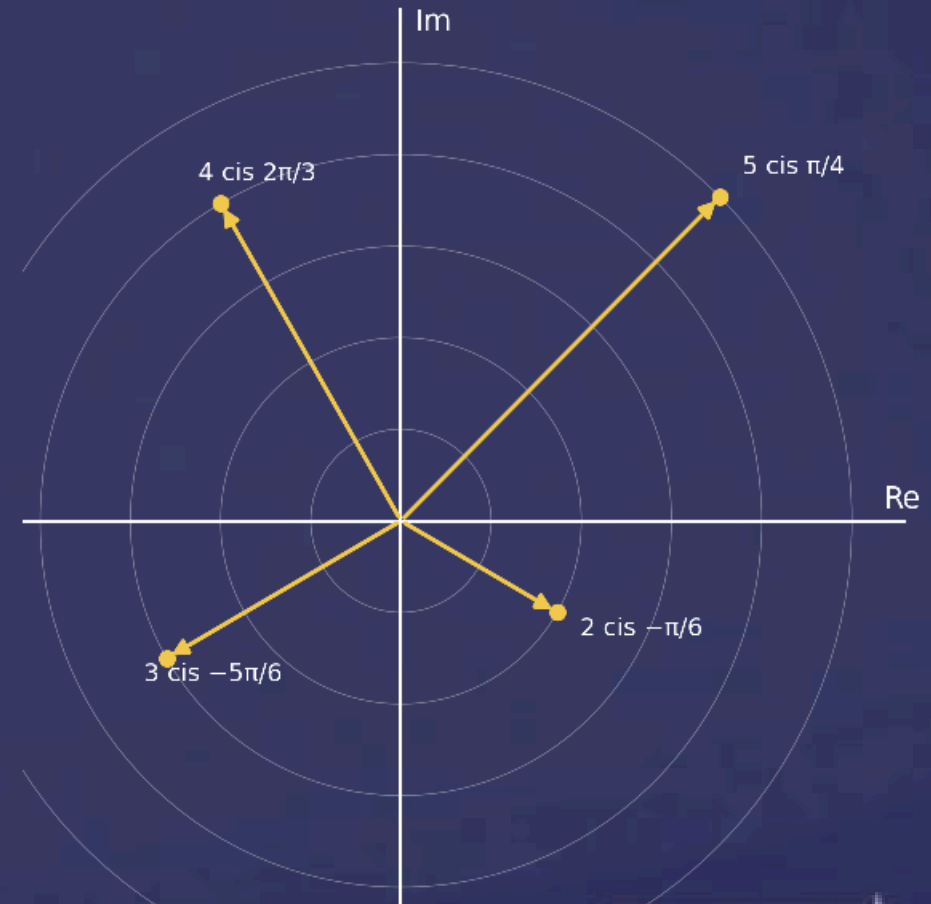
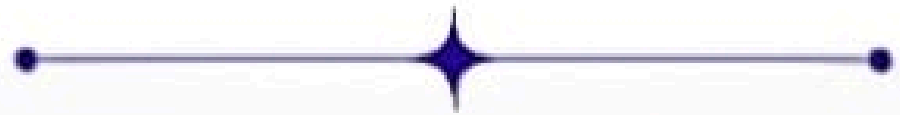
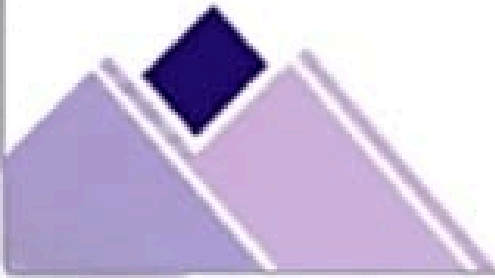
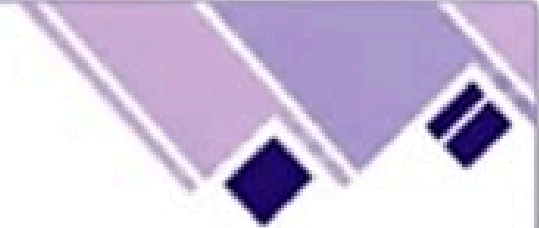
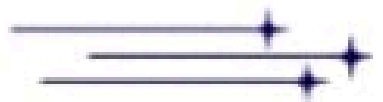


Fig 2.35



2.7.2 Euler's Form of the Complex Number

A third, compact way to write a complex number



Step 6 — Euler's formula

$$e^{i\theta} = \cos \theta + i \sin \theta$$

- ◆ This identity is called Euler's formula.
- ◆ It lets us rewrite the polar form using an exponential: $z = r e^{i\theta}$.
- ◆ This is called the Euler form of the complex number.
- ◆ Euler form is also convenient for multiplication, powers, and roots — same idea as polar form, written more compactly.



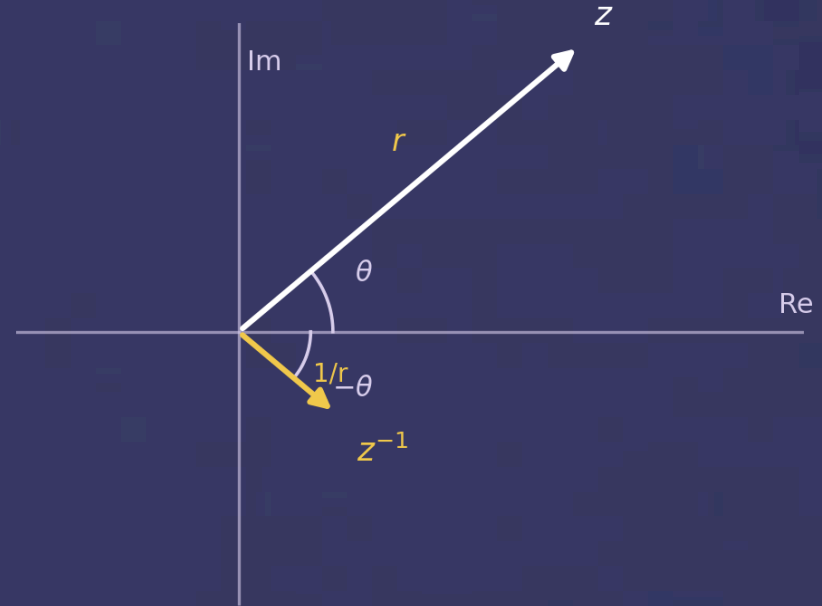
Property 1 — Reciprocal of z in Polar Form

If $z = r(\cos \theta + i \sin \theta)$, then $z^{-1} = (1/r)(\cos \theta - i \sin \theta)$.

Proof — step by step

1. $z^{-1} = 1/z = 1 / [r(\cos \theta + i \sin \theta)]$
2. Multiply top and bottom by the conjugate $(\cos \theta - i \sin \theta)$.
3. Denominator $\rightarrow r(\cos^2 \theta + \sin^2 \theta) = r \cdot 1 = r$.
4. So $z^{-1} = (1/r)(\cos \theta - i \sin \theta)$. ■

Geometrically: z^{-1} has modulus $1/r$ and argument $-\theta$ — same angle, flipped below the real axis, and scaled to length $1/r$.



z and z^{-1} as vectors from O

$$|z^{-1}| = 1/r, \quad \arg(z^{-1}) = -\theta$$

Property 2 – Product in Polar Form

If $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$, $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$, then $z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$.

Proof – step by step

1. $z_1 z_2 = r_1(\cos \theta_1 + i \sin \theta_1) \cdot r_2(\cos \theta_2 + i \sin \theta_2)$
2. $= r_1 r_2 [(\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2)]$
3. Recognise the angle-sum identities for $\cos(\theta_1 + \theta_2)$ and $\sin(\theta_1 + \theta_2)$.
4. $z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$. ■

So $\arg(z_1 z_2) = \arg z_1 + \arg z_2$ – moduli multiply, arguments add.



Angles add, giving the product vector $z_1 z_2$

$$|z_1 z_2| = r_1 r_2, \quad \arg(z_1 z_2) = \theta_1 + \theta_2$$

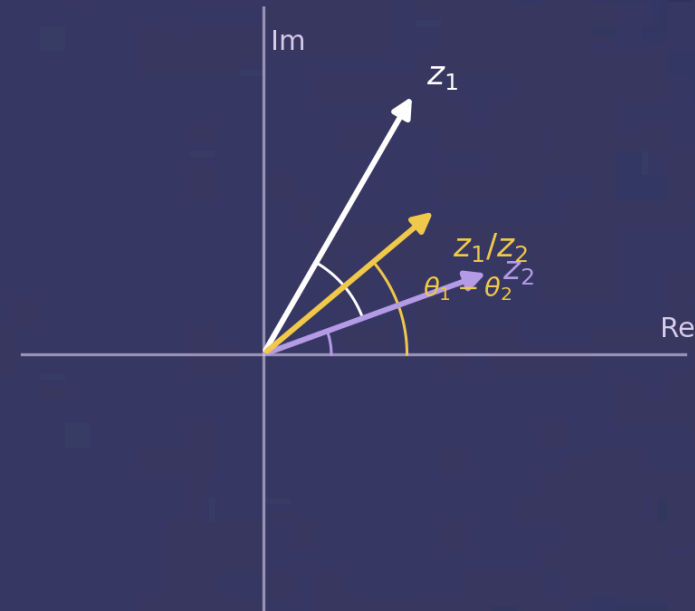
Property 3 – Quotient in Polar Form

If $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$, $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$, then $z_1/z_2 = (r_1/r_2)[\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$.

Proof – step by step

1. $z_1/z_2 = [r_1(\cos \theta_1 + i \sin \theta_1)] / [r_2(\cos \theta_2 + i \sin \theta_2)]$
2. Multiply top and bottom by the conjugate of the denominator, $(\cos \theta_2 - i \sin \theta_2)$.
3. Denominator $\rightarrow r_2(\cos^2 \theta_2 + \sin^2 \theta_2) = r_2$.
4. Numerator $\rightarrow r_1[(\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2 - \cos \theta_1 \sin \theta_2)]$
5. Apply angle-difference identities $\rightarrow z_1/z_2 = (r_1/r_2)[\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$. ■

So $\arg(z_1/z_2) = \arg z_1 - \arg z_2$ – moduli divide, arguments subtract.



Angles subtract, giving the quotient vector z_1/z_2

$$|z_1/z_2| = r_1/r_2, \quad \arg(z_1/z_2) = \theta_1 - \theta_2$$

Quick recap of Section 2.7

- ◆ Rectangular form: $z = x + iy \rightarrow$ best for addition/subtraction.
- ◆ Polar form: $z = r(\cos \theta + i \sin \theta) = r \operatorname{cis} \theta \rightarrow$ best for multiplication, powers, roots.
- ◆ $r = |z| = \sqrt{x^2 + y^2}$ and $\tan \theta = y/x$.
- ◆ $\arg z$ has infinitely many values; $\operatorname{Arg} z$ is the unique value with $-\pi < \theta \leq \pi$.
- ◆ Euler form: $z = r e^{i\theta}$, from $e^{i\theta} = \cos \theta + i \sin \theta$.

